Learning Management System(LMS)of the edX Website

Phase 1: Requirement Analysis

A. Who are the Users?

Example:

▼ **Super Admin**

- Full System Control
- Manage all Users
- Manage Roles and Permissions
- Manage Payment
- Manage Dashboard

▼ **Admin**

- Manage Courses
- Approve instructors
- Moderate Course Content
- Manage students

▼ **Instructor**

- Create Courses Content
- Update Course Content
- Upload Lessons

- Create Quizzes
- Create Assignments
- View student progress

▼ **Editor**

- Edit Course Content
- Approve or reject course before publish

▼ **Student**

- Register and login
- Enroll in Courses
- Watch Course
- Submit Assignments
- Browse Public Courses
- Get Certificate
- Give course review

▼ **Guest**

- Browse Public Courses
- Search Courses
- View Course Details
- Create account

B. Functional Requirements

Example:

▼ **Authentication**

- User Login/Logout
- User Registration
- Forgot Password
- Email Verification
- 2FA (Two-Factor Authentication)
- Change Password

▼ **RBAC(Role Based Access Control)**

- Role Management
- Permission Management
- Assign role to User
- Control access by role

▼ **Course System**

- Create Course
- Update Course
- Delete Course
- Publish/Unpublish Course
- Add course price
- Add course duration

▼ **Quiz System**

- Add MCQ Question
- Set Timer

- Auto Grading
- Show quiz result

▼ **Assignment System**

- Upload Assignment
- Grade Assignment
- Create Assignment
- Submit Assignment
- Track submission status

▼ **Category System**

- Multi-level Categories

▼ **Enrollments**

- Enroll in free /paid Course
- View Enrolled Courses

▼ **Payment System**

- Course Payment
- Verified Certificate Payment
- Payment history

▼ **Notifications**

- Course Update notification
- Assignment deadline notification

C. Non-Functional Requirements

▼ **Performance**

- Support 100k Users
- 10k Concurrent Users

▼ **Security**

- Password Hashing
- CSRF Protection
- XSS Protection
- SQL Injection Prevention
- Secure file upload

▼ **Scalability**

- Multiple Servers
- Cache Support

▼ **Availability**

- 99.9% uptime
- Backup System

▼ **Maintainability**

- Clean Architecture

▼ **Usability**

- Easy User Interface
- Mobile Responsive Design

D. Create SRS (Software Requirement Specification) Document

- Project Overview
- Features
- User Roles

- System Workflow
- Database Overview
- API Overview
- Security Rules
- Business Rules
- ER Diagram

System Design

▼ High Level Architecture

Client Side -Frontend(Blade/React/ Vue) - Backend API (PHP Laravel) - Services - Database - Cache - Stargat)

▼ Tech Stack

- Frontend(Blade or React)
- Backend(Laravel)
- Database(MySQL)
- Cache(Redis)
- Queue(Redis Queue)
- Storage(AWS S3)
- Realtime(Pusher/WebSocket)
- Server(Nginx)
- Deployment(Docker)
- CI/CD(GitHub Actions)

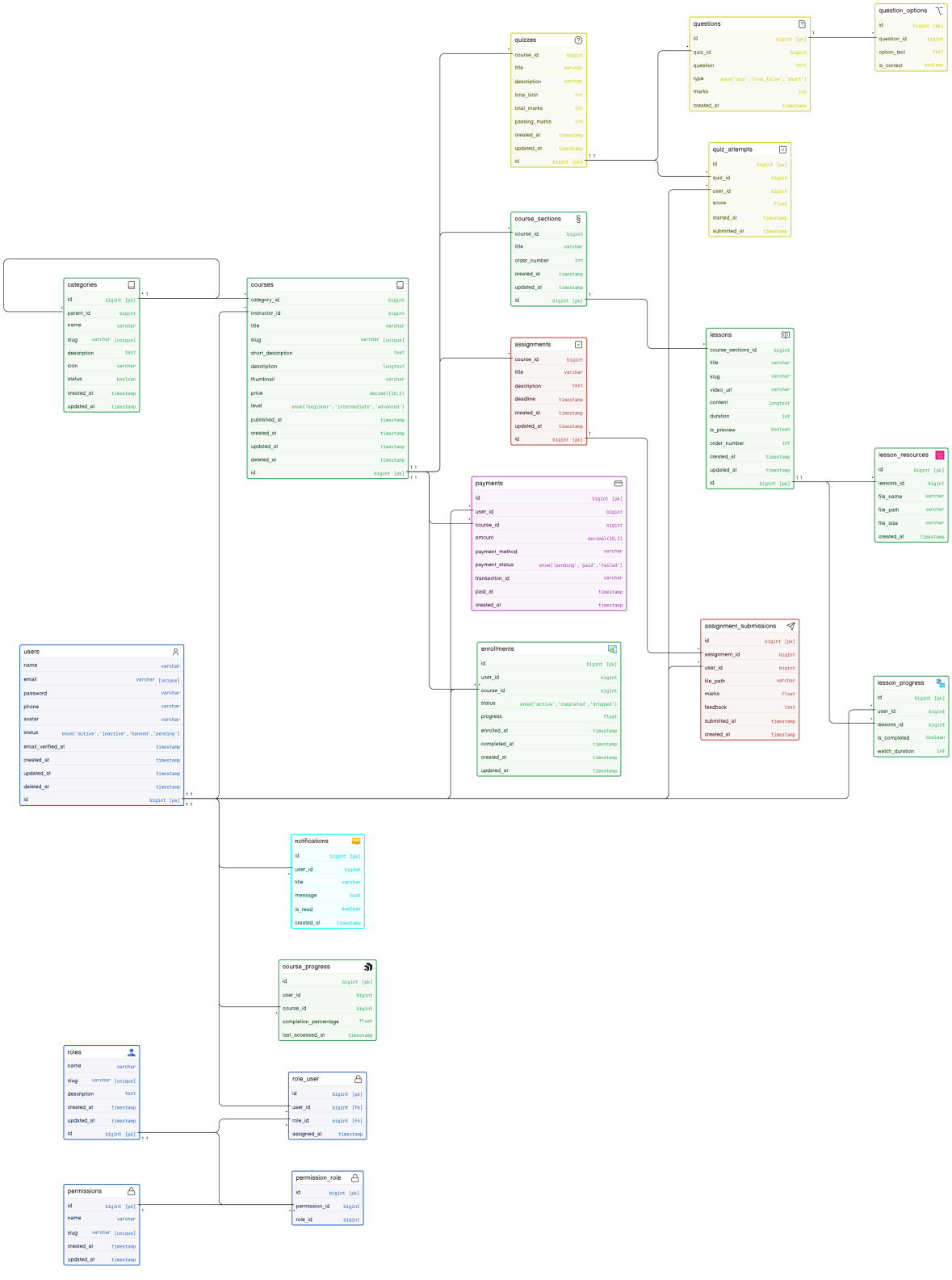
LMS Modules

1. Authentication
2. RBAC

3. User Management
4. Course System
5. Lesson Management
6. Enrollment System
7. Quiz System
8. Assignment system
9. Progress Tracking
10. Notification System
11. Payment System
12. Analytics

Entity Relationship Diagram

LMS Database Design of the edX Website


quizzes
course_id bigint
title varchar
description varchar
time_limit int
total_marks int
passing_marks int
created_at timestamp
updated_at timestamp
id bigint [pk]
questions
id bigint [pk]
quiz_id bigint
question text
type enum('mcq','true_false','short')
marks int
created_at timestamp
question_options
id bigint [pk]
question_id bigint
option_text text
is_correct boolean
quiz_attempts
id bigint [pk]
quiz_id bigint
user_id bigint
score float
started_at timestamp
submitted_at timestamp
course_sections
course_id bigint
title varchar
order_number int
created_at timestamp
updated_at timestamp
id bigint [pk]
categories
id bigint [pk]
parent_id bigint
name varchar
slug varchar [unique]
description text
icon varchar
status boolean
created_at timestamp
updated_at timestamp
courses
category_id bigint
instructor_id bigint
title varchar
slug varchar [unique]
short_description text
description longtext
thumbnail varchar
price decimal(10,2)
level enum('beginner','intermediate','advanced')
published_at timestamp
created_at timestamp
updated_at timestamp
deleted_at timestamp
id bigint [pk]
assignments
course_id bigint
title varchar
description text
deadline timestamp
created_at timestamp
updated_at timestamp
id bigint [pk]
lessons
course_sections_id bigint
title varchar
slug varchar
video_url varchar
content longtext
duration int
is_preview boolean
order_number int
created_at timestamp
updated_at timestamp
id bigint [pk]
lesson_resources
id bigint [pk]
lessons_id bigint
file_name varchar
file_path varchar
file_size varchar
created_at timestamp
payments
id bigint [pk]
user_id bigint
course_id bigint
amount decimal(10,2)
payment_method varchar
payment_status enum('pending','paid','failed')
transaction_id varchar
paid_at timestamp
created_at timestamp
assignment_submissions
id bigint [pk]
assignment_id bigint
user_id bigint
file_path varchar
marks float
feedback text
submitted_at timestamp
created_at timestamp
enrollments
id bigint [pk]
user_id bigint
course_id bigint
status enum('active','completed','dropped')
progress float
enrolled_at timestamp
completed_at timestamp
created_at timestamp
updated_at timestamp
lesson_progress
id bigint [pk]
user_id bigint
lessons_id bigint
is_completed boolean
watch_duration int
users
name varchar
email varchar [unique]
password varchar
phone varchar
avatar varchar
status enum('active','inactive','banned','pending')
email_verified_at timestamp
created_at timestamp
updated_at timestamp
deleted_at timestamp
id bigint [pk]
notifications
id bigint [pk]
user_id bigint
title varchar
message text
is_read boolean
created_at timestamp
course_progress
id bigint [pk]
user_id bigint
course_id bigint
completion_percentage float
last_accessed_at timestamp
roles
name varchar
slug varchar [unique]
description text
created_at timestamp
updated_at timestamp
id bigint [pk]
role_user
id bigint [pk]
user_id bigint [fk]
role_id bigint [fk]
assigned_at timestamp
permissions
id bigint [pk]
name varchar
slug varchar [unique]
created_at timestamp
updated_at timestamp
permission_role
id bigint [pk]
permission_id bigint
role_id bigint

eraser

eraser.io code link

https://app.eraser.io/workspace/exsUFzdGia6KUoKZVxBI?origin=share&diagram=SPmQSpAD6J5lhbmG6mcxV